Deliverable 1 of 7

Project Planning & Management

AWS Threat Detection, Monitoring, Incident Response & Forensics

DEPI – AWS Security Cloud Computing Track – Project 4

#	Team member	Role
1	Kareem Mohamed Abdalmged	Team Lead, IaC & Documentation
2	Adham Ramzy Mohamed Eltohamy	Threat Detection Engineer
3	Hossam Abdelghany Mostafa	Incident Response & Forensics Engineer
4	Alaa Ahmed Ramadan	Logging & Audit Engineer
5	Amira Ashraf Saber	Automation & Reporting Engineer

Instructor: Ahmed Gamal | Submission due: 16 October 2026 | Version 1.0

1. Project Proposal

A five-member team will design, build and defend a cloud security operations capability on AWS: a central logging and audit foundation, managed threat detection services, a rehearsed incident response and forensics process, and automated (SOAR-style) response with technical and executive reporting.

1.1 Problem Statement

Organisations run workloads on AWS (a public load balancer, web and application servers, a database and data buckets) without a complete audit trail or a rehearsed response process. Logs are disabled or scattered, findings from different services are not correlated, and when an instance, an access key or a bucket is compromised the team loses time deciding what to do and may destroy evidence while doing it. The project builds a reference security-operations environment that shows how native AWS services close these gaps: every action is logged, threats are detected and triaged, incidents are contained with preserved evidence, and repeatable responses are automated.

1.2 Objectives

1. Build a logging and audit foundation: a CloudTrail organization trail, CloudWatch Logs, AWS Config, VPC Flow Logs, S3 server access logs and ELB access logs delivered to S3 / CloudWatch with a retention policy and queried with Amazon Athena.
2. Enable threat detection with Amazon GuardDuty, AWS Security Hub, Amazon Detective, Amazon Inspector, Amazon Macie, AWS Trusted Advisor and AWS Config rules; triage sample findings and map them to MITRE ATT&CK.
3. Practise cloud incident response and forensics on three simulated incidents (compromised EC2 instance, exposed access keys, public S3 bucket): containment, evidence collection, timeline, root cause analysis and chain-of-custody notes.
4. Automate response with Amazon EventBridge, AWS Lambda, Systems Manager Automation and Security Hub custom actions in SOAR-style playbooks, and write technical and executive incident reports with lessons learned.
5. Deliver the full environment as Infrastructure as Code (Terraform) in a GitHub repository.

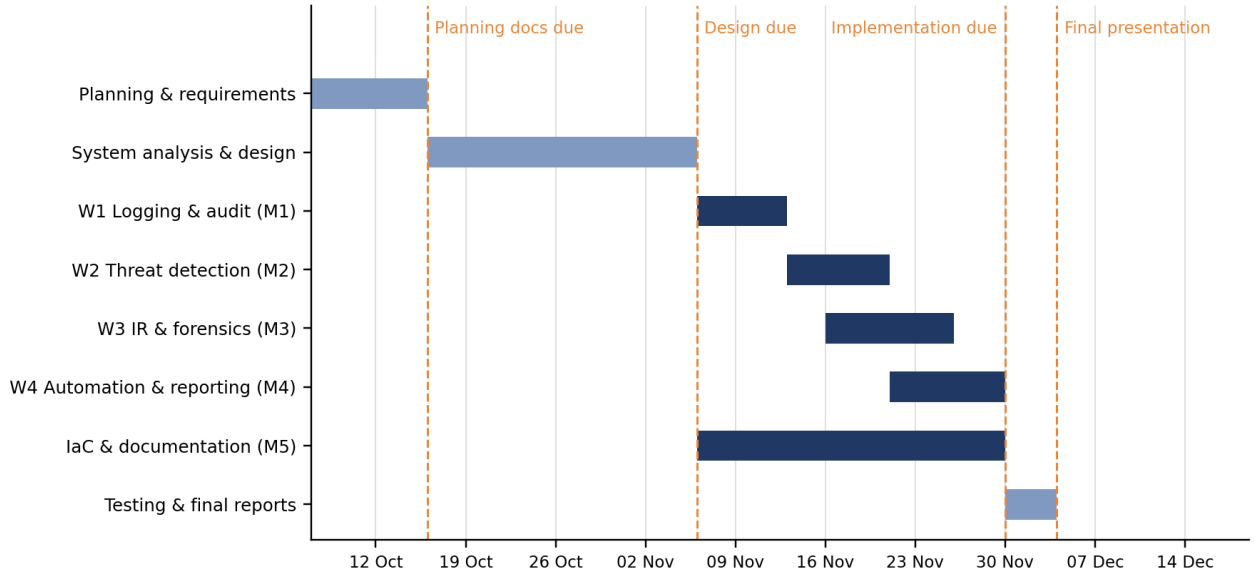
1.3 Scope

In scope	Out of scope (concepts and comparisons only)
Hands-on build of all services in the four project weeks, in one AWS region	Amazon Security Lake, CloudTrail Lake and third-party SIEM tools
Reference workload from the scenario: ALB, web EC2 (public subnet), app EC2 and RDS (private subnet), EBS, S3 data bucket, IAM users and access keys – lab scale with test data	Enterprise-scale multi-account and multi-region Security Hub / GuardDuty aggregation
Organization trail through AWS Organizations (shared team account as management account); single-account trail as fallback	Memory acquisition and live-forensics tooling on instances (compared conceptually)
Simulated incidents only: compromised EC2, exposed access keys, public S3 bucket; GuardDuty sample findings	Real malware, attacks on systems the team does not own, production workloads or real customer data
Architecture diagrams, log inventory, detection baseline, IR playbooks, forensic evidence, SOAR design and reports	Paid AWS Support plans (Trusted Advisor is limited to the checks available on the account's plan)
Terraform code, test reports and lab evidence on GitHub	Managed incident-response services and third-party SOC / MDR providers

2. Project Plan

The project follows the DEPI schedule: planning documents by 16 Oct, design by 6 Nov, implementation by 30 Nov and the final presentation on 4 Dec. The logging foundation (Week 1) starts first; the detection and incident response tracks start once the log sources are flowing, and the automation track starts once the first playbook is drafted.

2.1 Timeline (Gantt chart)



Dark bars = implementation tracks (owner in brackets). Light bars = documentation phases. Dashed lines = DEPI deadlines.

2.2 Milestones and Deliverables

Due date	Milestone	Deliverables
16 Oct 2026	Planning, literature review, requirements	Proposal, plan, roles, risk register, KPIs; literature review; stakeholder analysis, user stories, functional and non-functional requirements
6 Nov 2026	System analysis & design	Architecture diagrams (attack surface and assets, logging architecture, detection pipeline, IR and forensics flow, SOAR flow), log-archive layout, IAM role design, data-flow and sequence diagrams, technology stack
30 Nov 2026	Implementation	Terraform code on GitHub, four project weeks deployed with lab evidence: organization trail, Config recorder, Athena queries, detection baseline, IR playbooks and forensic evidence, EventBridge / Lambda demo
4 Dec 2026	Final presentation & reports	Test plan and results, final IR report and executive summary with lessons learned, user manual, presentation, optional demo video

2.3 Resource Allocation

- **People:** five members, one owner per project week plus a team lead (Section 3).
- **AWS accounts:** each member labs in a personal free-tier account; the integrated environment (including the AWS Organizations setup for the organization trail) is deployed in one shared team account for the final demo.
- **Budget:** USD 20 per account per month, enforced with AWS Budgets alerts at 50%, 80% and 100%. Detection services are enabled only in the week they are needed so that free trials (where offered) are used, then disabled.
- **Tools:** Terraform, GitHub (repository, issues, pull requests), draw.io for diagrams, Amazon Athena and CloudWatch for analysis, AWS CLI and Python (Lambda functions).

3. Task Assignment & Roles

Each member owns one project week end to end: build, evidence, documentation and demo. Member 5 leads the team and owns the IaC and shared documents.

Member	Role	Owns	Key deliverables
M1 Alaa Ahmed Ramadan	Logging & Audit Engineer	Week 1 – Logging & Audit Foundation	Log architecture diagram, CloudTrail org trail, Config recorder, VPC Flow / S3 access / ELB logs, log retention policy, Athena queries, audit log inventory
M2 Adham Ramzy Mohamed Eltohamy	Threat Detection Engineer	Week 2 – Threat Detection Services	Detection baseline, GuardDuty / Security Hub findings, Config rules, Inspector scan, Macie, Detective and Trusted Advisor results, triage notes, MITRE ATT&CK mapping
M3 Hossam Abdelghany Mostafa	Incident Response & Forensics Engineer	Week 3 – Cloud Incident Response & Forensics	IR playbooks (EC2, access keys, S3), containment checklist, forensic evidence collection, incident timeline, root cause analysis, chain-of-custody notes
M4 Amira Ashraf Saber	Automation & Reporting Engineer	Week 4 – Automation & Executive Reporting	Automated response playbook, EventBridge / Lambda / SSM Automation demo, Security Hub custom actions, SOAR design, technical and executive IR reports, lessons learned
M5 Kareem Mohamed Abdalmged	Team Lead, IaC & Documentation	Cross-cutting	Terraform modules and GitHub repo, AWS Organizations and Budgets, all diagrams, DEPI documentation set, final report and presentation

Shared responsibilities

- Every member writes the Terraform module and README section for their own week; M5 reviews and merges pull requests.
- Every member destroys or disables billable resources after each lab session.
- Simulations run only on tagged lab resources in an isolated test subnet; no real attacks and no real data.
- Every member presents their own part in the final presentation.
- Backup owners: M1 ↔ M2, M3 ↔ M4; M5 covers documentation for anyone unavailable.

4. Risk Assessment & Mitigation Plan

The highest risk is unexpected AWS charges from services that bill by usage or after a free trial (Config, GuardDuty, Detective, Macie, Inspector, CloudWatch Logs, Athena scans); it is controlled with budgets, staged enablement and teardown after every session.

#	Risk	Likelihood	Impact	Mitigation	Owner
R1	AWS bill overrun (Config recorder, GuardDuty, Detective, Macie, Inspector after free trial; CloudTrail data events; CloudWatch Logs ingestion; Athena scans)	High	High	Budgets alerts; enable each service only in its week; S3 lifecycle rules; scope Macie to one test bucket; limit Athena scans with partitions and workgroup limits; destroy / disable after every session	M5
R2	Access keys leaked to GitHub (including the keys used in the exposed-key simulation)	Medium	High	Simulation uses a canary IAM user with minimal permissions and dummy keys; no long-lived keys in code; .gitignore for tfstate / tfvars; GitHub secret scanning	M5
R3	Simulation affects other resources or breaches AWS usage policy	Low	High	Simulate only on tagged test resources in an isolated subnet; use GuardDuty sample findings; no attacks on systems the team does not own	M3
R4	Organization trail not possible (AWS Organizations or management-account permissions unavailable)	Medium	Medium	Set up Organizations early in the shared account; fallback: multi-region trail in a single account, documented	M5
R5	Weeks 2–4 blocked waiting for logging foundation or for detection data	Medium	Medium	Deliver logging as a reusable Terraform module first; enable GuardDuty early in Week 2; use sample findings	M1
R6	A member becomes unavailable	Medium	Medium	Work documented in the repo as it is built; backup owner per week	M5
R7	Service quotas, free-trial eligibility or regional availability limits on new accounts	Low	Medium	Check Service Quotas and service availability during design; request increases early	All
R8	Documentation lags behind implementation	Medium	High	Capture screenshots, queries and outputs during each lab; weekly documentation check	M5
R9	Evidence integrity or chain-of-custody gaps	Medium	Medium	Evidence bucket with versioning and restricted access; SHA-256 hashes; custody log filled at every hand-over	M3

#	Risk	Likelihood	Impact	Mitigation	Owner
R10	Automated response isolates the wrong resource (false positive)	Medium	Medium	Scope automation by resource tag; test in the isolated subnet; analyst-approved custom action as alternative; SNS notification for each action	M4

5. Key Performance Indicators (KPIs)

The project succeeds when every control is proven by a test, incidents are detected and contained within target times with preserved evidence, the environment rebuilds from code, and costs stay within budget.

KPI	Target	How measured
CloudTrail organization trail covering all accounts and regions with log file validation	100%	CloudTrail console / CLI; digest validation
Log sources delivered centrally (CloudTrail, CloudWatch Logs, Config, VPC Flow Logs, S3 access logs, ELB access logs)	6 of 6	Audit log inventory; objects present in the log archive
Log retention policy applied to log buckets and log groups	100%, ≥ 30 days	S3 lifecycle rules; CloudWatch retention settings
Saved Athena queries returning correct results (API activity, rejected traffic, S3 access, ELB errors)	≥ 8 queries	Query results checked against known test events
Detection services enabled and reporting (GuardDuty, Security Hub, Detective, Inspector, Macie, Trusted Advisor, Config rules)	7 of 7	Service consoles; Security Hub finding sources
Triaged findings mapped to MITRE ATT&CK techniques	100% (≥ 10 findings)	Triage notes and mapping table
Time to detect simulated incident	< 15 minutes	Simulation timestamp vs finding / alarm time
Time to contain a simulated incident manually (isolate EC2, revoke keys, block public access)	< 10 minutes	IR playbook timeline
Time to contain automatically after the finding event	< 2 minutes	EventBridge / Lambda execution logs
Incidents with EBS snapshot / evidence hashed and chain-of-custody entry	100%	Evidence bucket; custody log
IR playbooks tested end to end (EC2, access keys, S3)	3 of 3	Playbook test records
Lab resources defined as code	100%	Full rebuild with terraform apply in a clean account
AWS spend	≤ USD 20 / account / month	AWS Budgets / Cost Explorer
DEPI milestones delivered on time	100%	Milestone checklist